

Exploring the Trade-offs between Energy and Performance of Federated Learning Algorithms

1 April 2024 - 9 July 2026

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This report presents my activity as part of my thesis, directed by **Prof. Georges Da Costa** and **Assoc. Prof. Millian Poquet**, within the **SEPIA team**.

Contents

1	Introduction	3
1.1	Thesis context	3
1.2	Thesis initial objectives	3
2	Thesis Progress	3
2.1	1st year	3
2.2	2nd year	3
2.3	3rd year - current year	4
2.3.1	Towards Sustainable Federated Learning: A Survey on Energy and Experimental Practices	4
2.3.2	FedE-ator: A Reproducible Framework for Studying Energy Consumption in Federated Learning Systems	5
2.4	Next year	6
3	Work situation	7
3.1	Difficulties encountered	7
3.2	Research Outputs	7
3.3	Teaching	7
3.4	Training progress	7

1. Introduction

1.1. Thesis context

Federated Learning (FL) is an emerging distributed ML paradigm where nodes compute local gradients and share them with a central server that coordinates training. This approach allows nodes (*e.g.*, IoT devices, smartphones) to train models collaboratively while keeping data local, thus addressing privacy and legal concerns.

However, FL introduces new challenges beyond traditional ML, including: (i) communication and energy overhead, (ii) longer training due to non-iid data, and (iii) unbalanced workloads across heterogeneous devices. Energy efficiency remains largely overlooked in recent FL variants, including cross-device FL, cross-silo FL, federated transfer learning, and personalized FL.

The ANR project DELIGHT¹ (started in early 2023) aims to integrate energy efficiency as a key metric in FL to promote sustainable learning. This position will address the first challenge using the Flower framework.

1.2. Thesis initial objectives

Our team contributes to the DELIGHT project by modeling energy consumption, optimizing hybrid CPU/GPU usage, and developing middleware for clusters, mesh, and cloud environments. [SDSE20]

My PhD focuses on exploring energy-performance trade-offs in FL, proposing new energy estimation methods, and developing models and libraries to measure energy at various levels (data, hardware, software, network). I aim to leverage framework parameters like floating-point precision and data transfer strategies, with use cases in speech recognition or object detection.

Phases of the PhD:

- Set up an experimental environment on Grid'5000 (g5k) to gather performance and energy metric.
 - Create a use case for the Flower framework.
 - Build a reproducible and automated framework to obtain metrics for this use case.
- Propose, model, and implement the different leverages.
- Explore the impact of leverage on both energy and performance.

2. Thesis Progress

2.1. 1st year

From April 2024 to the end of August 2024, I:

- Conducted a survey of related work (30 articles).
- Carried out initial experiments on energy measurement in a distributed system (Grid'5000).

2.2. 2nd year

From August 2024 to August 2025, I:

- Prepared a survey paper and revised it for submission.
- Developed FedE-ator (version 0), a framework for measuring energy consumption of FL in distributed systems.

Framework functions:

- DVFS (controlling CPU/GPU frequencies to analyze their impact on FL performance and energy consumption)
- Multiple instances and repetition in one run.
- Configuration for FL framework flexibility.

¹<https://rachid-elazouzi.github.io/DELIGHT/>

Framework limitations:

- Works only on CPU or only GPU (does not support two types of hardware in one experiment).
- Experiment: only validated within the Flower framework.

At the end of the second year, I declared the plan for the upcoming years as shown in Table 1, details of progress will be discussed in Section 2.3.

Table 1: Current progress on the objectives defined at the end of the second year.

Planned objectives (end of second year)	Current status
Framework heterogeneity: Develop a heterogeneous framework supporting both CPU and GPU nodes within the same experiment.	Completed
Framework reproducibility: Package the environment with Nix, allowing users to reproduce experiments and results reliably.	In progress (50%), Documentation completed; Nix packaging pending.
Scientific evaluation: Extend experiments with various FL frameworks and settings to investigate the trade-off between accuracy and energy consumption.	Completed
Resource optimization: Investigate real-world optimization and resource allocation by integrating optimization algorithms into the FL system together with energy measurement.	Target refined (Section 2.4), In progress (50%)
Publications: Submit and publish research papers.	In progress (50%), 1 workshop paper accepted, 1 journal paper under second-round review.

2.3. 3rd year - current year

From September 2025 to July 2026, I focused on two parts: revised and submitted survey paper and improved FedE-ator framework.

2.3.1. Towards Sustainable Federated Learning: A Survey on Energy and Experimental Practices

This work presents a bibliographic survey of energy-related studies in FL published between 2020 and 2024. From 263 screened articles, I selected 50 representative works based on multiple relevance criteria to analyze the landscape of energy-aware FL. The analysis focuses on how energy aspects are addressed and evaluated in the literature, with particular attention to experimental practices. Specifically, I investigate what experimental methods are used, how performance-energy tradeoffs are studied, what resources are considered, what simulation or real frameworks are employed, and how energy is measured.

Despite recent advances, challenges remain in designing real-world experiments, standardizing energy modeling, and addressing carbon emissions, fairness, and performance-energy tradeoffs. I proposed a Reproducibility Assessment Framework (RepS) quantifying reproducibility across five dimensions, revealing that 80% of selected papers fall within the low reproducibility range. I also proposed a unified evaluation framework built around three composite trade-off metrics between energy-time-accuracy, to support rigorous analysis of the energy-performance tradeoff.

Currently, this survey is submitted to *Sustainable Computing: Informatics and Systems Journal*², the timeline is:

- First submission on 29/08/2025
- Reviewer’s feedback *main revision*, on 14/05/2026
- Revised and re-submission on 11/06/2026

²<https://www.sciencedirect.com/journal/sustainable-computing-informatics-and-systems>

2.3.2. FedE-ator: A Reproducible Framework for Studying Energy Consumption in Federated Learning Systems

As planned, I have further extended FedE-ator to Version 2, adding support for heterogeneous hardware and multiple FL frameworks. The updated architecture and capabilities are presented as following.

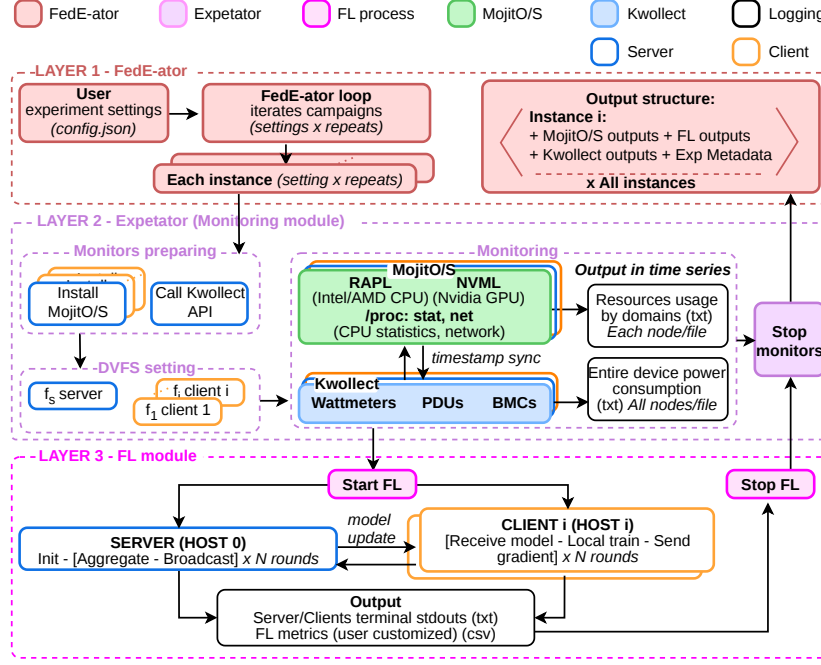


Figure 1: FedE-ator workflow (Expetator, MojitO/S, Kwllect as 3rd-party tools.)

Figure 1 illustrates the workflow of the FedE-ator framework for a campaign, showing the operations executed on each host (server and clients) from the start of the experiment until completion, including both FL execution and system monitoring and logging processes. To support running multiple instances, FedE-ator provides a configuration file storing all required settings for each experiment instance. FedE-ator sequentially reads these configurations, executes each sub-experiment, and stores the resulting outputs in a predefined directory following a consistent structure. FedE-ator is developed for Linux-based distributed systems. To support simultaneous monitoring of both CPU and GPU within a single experiment, this version of FedE-ator relies on a patched version of Expetator from a dedicated branch³ instead of the main installation package. This branch enables adaptive monitoring in one run by selecting appropriate monitoring methods according to the hardware configuration of each node (CPU or GPU).

The experiments were conducted on Grid'5000⁴ platform. The summary of the experiment settings is shown in Table 2. The experimental evaluation consists of three scenarios, with repetitions when feasible to reduce variability and limit the carbon footprint of large-scale executions:

SC1-Heterogeneous: validates FedE-ator across heterogeneous CPU/GPU nodes under three [CPU, GPU] frequency settings (2 runs per configuration).

SC2-Large-scale: validates framework's stability at scale (90 nodes) under $f_{\min}$ and $f_{\max}$ (1 run per configuration).

SC3-Cross-framework analysis: evaluates energy-performance tradeoff across two FL frameworks, three models and four CPU frequencies (5 runs per configuration).

Due to the space limitation of this report, Fig. 2 is selected from the results of SC3 to illustrate the energy consumption required to reach different target accuracies under various CPU frequencies, ML models, and FL frameworks. The results show that an intermediate CPU frequency (1.8 GHz

³<https://gitlab.irit.fr/sepia-pub/expetator/-/tree/Mai>

⁴<https://www.grid5000.fr/>

Table 2: Summary of Experimental Scenarios and Settings

Category	SC1-Heterogeneous	SC2-Large-scale	SC3-Cross-framework analysis
Server	nova-10	gros-104	nova-14
Clients	gemini-[1,2], nova-[11,12]	89 nodes (gros-[])	nova-[15,16,17]
Freq.*(GHz)	$f_{\min}$: [0.135, 1.2] f_{int} : [0.832, 1.7] $f_{\max}$: [1.53, 2.1]	$f_{\max}$: 2.2	$f_{\min}$: 1.2 f_{int} : {1.5, 1.8} $f_{\max}$: 2.1
Framework	Flower	Flower	Flower [BTM+20], Fed-Scale [LDS+22]
Library	PyTorch	TensorFlow	PyTorch
Model	CustomCNN	CustomCNN	3 models**
Epochs/Rounds	10/3	1/5	1, 512*** /10

* In SC1, frequencies are represented as a set of [CPU, GPU]. In SC2 and SC3, only CPU frequency is applied. $f_{\min}$ and $f_{\max}$ are the minimum and maximum frequencies of the CPU/GPU, f_{int} are the available frequencies between them.

** CustomCNN (LeNet5), MobileNetV2, ResNet18.

***1 Flower epoch $\approx$ FedScale 512 local steps (CIFAR-10, 16,667 samples, batch size 32).

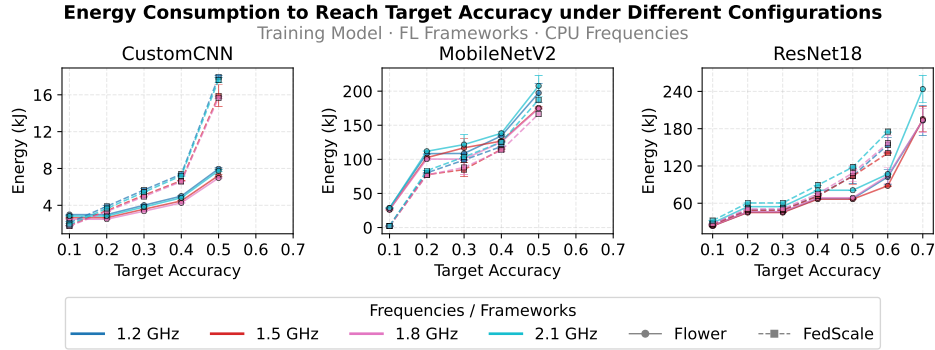


Figure 2: Energy consumption to reach target accuracy (within 10 rounds) across CPU frequencies, models, and FL frameworks (SC3). Each point represents the mean energy over 5 repeated runs, error bars indicate standard deviation.

for CustomCNN/MobileNetV2 and 1.5 GHz for ResNet18) provides the best energy efficiency, while the highest frequency increases energy consumption due to disproportionate power growth. Model complexity also strongly affects energy usage: larger models generally require more energy, although MobileNetV2 consumes more than ResNet18 because it is optimized for mobile inference rather than server-side training. The comparison between FL frameworks reveals a model-dependent impact, with Flower being more energy-efficient for CustomCNN and ResNet18 but less efficient for MobileNetV2. Finally, energy increases non-linearly with target accuracy, showing diminishing returns at higher accuracy levels, while discrete accuracy jumps between thresholds reflect the round-based nature of FL evaluation.

2.4. Next year

My plan for the last year includes the following:

- Extending FedE-ator with online DVFS-based energy efficiency for adaptive resource management (target: finish in September 2026)
 - Phase-aware adaptation
 - Round-time-aware adaptation
 - CPU-utilization-aware adaptation (load balance in system)
- Thesis preparation: start in September 2026.

3. Work situation

3.1. Difficulties encountered

There have been no major difficulties, and my work is progressing as planned.

3.2. Research Outputs

- The **FedEator** framework (version 2), which is available as open-source here: <https://gitlab.irit.fr/sepia-pub/delight/eflwr>. With this version, FedE-ator is able to manage heterogeneous HW (CPUs and GPUs) in one run. It also allows different FL frameworks (such as Flower and FedScale).
- Paper describing the **FedE-ator** (*FedE-ator: A Reproducible Framework for Studying Energy Consumption in Federated Learning Systems*) was accepted to the PECS workshop⁵ in the EUROPAR conference.
- Survey paper (*Towards Sustainable Federated Learning: A Survey on Energy and Experimental Practices*) was submitted to the SUSCOM Journal and is waiting for second-round review after a major revision decision for the first round.

3.3. Teaching

I participated in teaching and giving tutorials for TP and TD sessions of the CSA class (for international M1 students, class in English), specifically on **monitoring** and **MPI** in the *Parallel* course (total 26 hours).

3.4. Training progress

Table. 3 shows the total training courses that I took during 3 years.

Table 3: Completed doctoral training activities.

Category	Training activity	Date	Hours
Quality of life at work	<i>Débat théâtral "Entre les lignes"</i>	Apr. 2025	3
Dissemination of knowledge	<i>Journée Doctorants ASR</i>	Jun. 2024	8
	<i>MOOC: Reproducible Research II: Practices and Tools for Managing Computations and Data</i> (Inria)	Sep. 2024	10
Ethics and Scientific Integrity	<i>MOOC: Éthique de la recherche</i> (Université de Lyon)	Apr. 2026	6
	<i>MOOC: Intégrité scientifique dans les métiers de la recherche</i> (Université de Bordeaux)	Aug. 2025	6
	<i>MOOC: Recherche reproductible : principes méthodologiques pour une science transparente</i>	Jul. 2024	6
International / Languages	<i>Cours de Français Langue Étrangère (FLE)</i> (Grade: 14/20)	Jun. 2025	23
Interdisciplinarity / Scientific	<i>EcoICT Fall School 2024</i>	Oct. 2024	20
	<i>Summer School Green IT 2026</i>	Jun. 2026	34
	<i>What do we mean by Digital Sobriety?</i>	Oct. 2024	2
Career progression	<i>PhDOOC: PhD and Career Development</i>	Mar. 2025	12
Total			130

⁵<https://pecs-workshop.github.io/2026/>

References

- [BTM⁺20] Daniel J Beutel, Taner Topal, Akhil Mathur, Xinchu Qiu, Javier Fernandez-Marques, Yan Gao, Lorenzo Sani, Kwing Hei Li, Titouan Parcollet, Pedro Porto Buarque de Gusmão, et al. Flower: A friendly federated learning research framework. *arXiv preprint arXiv:2007.14390*, 2020.
- [LDS⁺22] Fan Lai, Yinwei Dai, Sanjay S. Singapuram, Jiachen Liu, Xiangfeng Zhu, Harsha V. Madhyastha, and Mosharaf Chowdhury. FedScale: Benchmarking model and system performance of federated learning at scale. In *International Conference on Machine Learning (ICML)*, 2022.
- [SDSE20] Roy Schwartz, Jesse Dodge, Noah A Smith, and Oren Etzioni. Green AI. *Communications of the ACM*, 63(12):54–63, 2020.